ABDUL SALAM KAREEM

Kasaragod, Kerala, India | +91-8078141973 | salamkareemk@gmail.com | linkedin.com/in/abdul-salam-kareem

PROFESSIONAL SUMMARY

AI Engineer and Python Full Stack Developer with hands-on experience building Django and Flask applications, REST APIs, and AI-powered systems. Skilled in Generative AI, LLMs, and Retrieval-Augmented Generation (RAG), including document chunking, embedding generation, vector database indexing, and semantic search. Experienced in developing deep learning models for NLP and computer vision with TensorFlow, PyTorch, and Scikit-learn, and in shipping features end to end from data preprocessing to deployed web application. B.Tech graduate in Artificial Intelligence and Machine Learning with strong analytical and problem-solving skills and a track record of delivering working products in internship and project settings.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, HTML, CSS

Backend & APIs: Django, Flask, Node.js, Express.js, REST APIs, API Integration

Generative AI & NLP: LLMs, Retrieval-Augmented Generation (RAG), Embeddings, Semantic Search, Prompt Engineering, Document Chunking, NLP

Data & ML Libraries: TensorFlow, PyTorch, Scikit-learn, OpenCV, Pandas, NumPy, YOLO, LSTM, BERT

Web Development: MERN Stack (MongoDB, Express.js, React.js, Node.js), Vite, React Router, Django Templating, HTML, CSS

Databases: MongoDB, MySQL, SQL, Vector Databases

Tools & Platforms: Git, GitHub, VS Code, Vercel, Microsoft Azure AI

EXPERIENCE

Full Stack Developer – MERN | HACA – Haris & Co., Kozhikode Sep 2026 – Present

- Developed a placement application on the MERN stack (MongoDB, Express.js, React.js, Node.js) to help students discover relevant job opportunities.
- Integrated APIs to fetch and use data within the application, building both frontend and backend features.
- Worked on AI-related features and explored practical applications of AI within the application.
- Translated application requirements into practical features, solving real-world development challenges through hands-on project work.

Advanced Python Django & GenAI Intern | HACA – Haris & Co., Kozhikode Dec 2025 – Aug 2026

- Develop backend systems in Python and Django, applying structured project architecture, reusable app design, and software development best practices.
- Build AI-integrated web applications that connect Django backends to LLM-based workflows for real-time, context-aware responses.
- Implemented document chunking, embedding generation, semantic retrieval, and chatbot integration for an institutional AI assistant.
- Translate Generative AI concepts into production-style features through hands-on development, code review, and iterative testing.

AI Intern | SmecLabs, Kochi May 2024 – Jun 2024

- Developed a hate speech detection web application using deep learning for automated text classification and content moderation.
- Integrated an LSTM-based model with a web interface to serve real-time inference on user-submitted text.
- Improved model performance through text preprocessing, dataset refinement, and systematic evaluation of accuracy and loss metrics.

AI and Python Intern | Codsoft (Remote) Jun 2024 – Jul 2024

- Built multiple Python applications, including an NLP-powered AI chatbot and an interactive game, delivered against internship milestones.
- Applied NLP concepts to enable intelligent conversational behavior and user interaction in the chatbot.
- Strengthened debugging, logical reasoning, and clean code practices across independently delivered modules.

PROJECTS

HACAGPT – RAG Based AI Chatbot | Internship Project | GitHub 2026

- Designed and developed an institutional AI chatbot using LLMs and Retrieval-Augmented Generation to answer academic and institutional queries from HACA's knowledge base.
- Implemented the complete RAG pipeline in Python: data collection, preprocessing, document chunking, embedding generation, vector database indexing, semantic similarity search, and LLM response generation.
- Improved retrieval accuracy by tuning chunking strategies and retrieval parameters, and served the system through an interactive Flask web application.
- Technologies: Python, LLM, RAG, Flask, Vector Database, Embeddings, NLP.

MovieWorld – React Movie Database | Personal Project | GitHub 2026

- Built a responsive movie discovery application consuming the TMDB API for trending, popular, top-rated, upcoming, and latest titles.
- Implemented search, genre-based filtering, detailed movie pages, and persistent Favorites and Watched lists using Local Storage.
- Structured the UI with reusable React components and deployed the application to Vercel.
- Technologies: React.js, Vite, JavaScript, React Router, TMDB API, HTML, CSS, Vercel.

Traffic Sign Detection and Vehicle Counting System | Academic Project Nov 2024 – Apr 2025

- Developed a real-time traffic monitoring and analysis system using deep learning models for traffic sign detection and vehicle counting.
- Improved automation in smart traffic analysis using object detection and computer vision with OpenCV, TensorFlow and YOLO, served through a Flask web interface.
- Served results through a Flask web interface and evaluated the models using accuracy, precision, and recall.
- Technologies: Python, OpenCV, TensorFlow, YOLO, Flask.

Event Management and Certificate Generation System | Django Project | GitHub 2025

- Built a full-stack event management platform in Django covering participant registration, event handling, and administration.
- Implemented automated, dynamic certificate generation for participants, removing manual effort from event coordination.
- Technologies: Django, Python, HTML, CSS.

Hate Speech Detection Website | Internship Project 2024

- Built a deep learning moderation system using LSTM and BERT models to identify and classify harmful content in user-generated text.
- Applied supervised learning, text preprocessing, and comparative model evaluation to select the best-performing classifier.
- Technologies: Python, NLP, Deep Learning, LSTM, BERT.

EDUCATION

B.Tech in Artificial Intelligence and Machine Learning

Srinivas University Institute of Engineering and Technology, Mangalore | Aug 2021 – May 2025

CERTIFICATIONS

- IT Specialist – Python | Certiport
- IT Specialist – JavaScript | Certiport
- Career Essentials in Generative AI | Microsoft and LinkedIn
- Artificial Intelligence Internship | Nexel (Futureize The Innovation)
- Object Oriented Programming using Python | Infosys Springboard
- Machine Learning Crash Course | Great Learning
- Game Development using Pygame and ChatGPT for Everyone | HCL GUVI (partnered with Google)
- Workshop on Advanced AI-ML Programming using Open-Source Tools | Kakunje Software Pvt. Ltd.

LANGUAGES

English, Malayalam, Hindi